

Support for the New Claims

The overall architecture recited in new independent claims 19 and 35 is summarized at page 8, lines 9-26, and is described in detail with reference to FIG. 1 at pages 13-15, in particular from page 14, line 2, through page 15. These passages describe the mate creation component 110, the transcoding components 120, the manifest files 121 pointing to unique interleaved combinations of chunks 113 of the ensemble 112, the ledger 122, the detection component 130 running a camera cuts detection algorithm 131 that devises time codes from the pirate webcasting and builds reconstructed manifest files 121', and the retrieval component 140 searching the ledger for manifest files equal to the reconstructed manifest files 121'. Original claims 1, 10, 11, 14 and 15 recite this architecture in claim form.

As to the production side, the structured list of edit instructions with per-cut out-point and in-point time codes and source-camera identifiers is described at page 16, lines 18-28, and at page 18, lines 13-17. Example 2 at pages 33-35, with Tables 1 and 2, sets out the source camera and the in-point and out-point time codes of each cut and describes extending the duration of Cut 2 by 10 frames so that the start of Cut 3 is delayed by 10 frames while synchronization is maintained from Cut 4 onwards. Example 2 thereby also supports new claims 20 and 21, in which the mate transition occurs at a later timing through extension of the selection of the first camera. Variation at a plurality of selected camera cuts and manifest files representing combinations of per-cut timing choices are supported at page 16, lines 1-17, where each further camera cut doubles the number of content versions and manifest files.

As to distribution and the record of associations, segmentation into chunks, delivery through a content delivery network with adaptive streaming, the generation of manifest files and the ledger identifying which end user received which manifest file appear at page 8, lines 9-21, and at page 14, lines 10-21. The mixing component integrating reference and mate chunks is described at page 18, lines 23-28. The progressive association of each user with a distinct manifest file is described at page 40, lines 5-7. Unicast delivery appears at page 30, lines 18-20. See also original claims 6 and 11-14.

As to the detection side, including monitor-side method claim 38, the operations of receiving the suspected unauthorized distribution, applying a scene-change detection algorithm to identify time codes of camera cuts, building one or more reconstructed manifest files from the identified time codes and searching the ledger to identify the associated recipient are described at page 8, lines 15-21, at page 14, line 22, through page 15, and in method form at page 30, lines 1-10, describing the ledger recording step 270, the monitoring step 280 detecting scene changes and devising time codes and the searching step 290. Original claims 15, 16 and 17 recite the same operations in claim form.

As to the remaining dependent claims, the perceptual hash and fuzzy matching comparisons of new claims 25, 26, 36 and 37 are described at page 30, lines 21-28, in Example 5 at pages 40-43 and in original claims 2 and 3. Probabilistic Tardos fingerprinting for collusion scenarios, new claims 29 and 30, is described at pages 24-26 and in original claim 5. The overlay of additional audio or video elements, new claim 34, is described at page 24, lines 12-16, at page 30, lines 11-13, and in original claim 4. The Edit Decision List implementation of the structured list, new claim 22, is described at page 16, lines 18-25. Capture of live video from diverse viewpoints, new claim 23, is recited in original claim 16. The computing environment recited in the preambles is described at page 18, lines 1-3, and in original claim 8.